



# Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

|                     |                     |                     |                                                             |
|---------------------|---------------------|---------------------|-------------------------------------------------------------|
| <b>Class:</b>       | <b>TE-AI&amp;DS</b> | <b>Semester:</b>    | <b>V</b>                                                    |
| <b>Course Code:</b> | <b>2015111</b>      | <b>Course Name:</b> | <b>Statistics for Machine Learning and Data Science Lab</b> |

|                                 |                                                                                                 |
|---------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Name of Student:</b>         | <b>Yash Vilas Karande</b>                                                                       |
| <b>Roll No. :</b>               | <b>17</b>                                                                                       |
| <b>Experiment No.:</b>          | <b>4</b>                                                                                        |
| <b>Title of the Experiment:</b> | <b>Perform Statistical Inference, Estimation, and Hypothesis Testing on Real-World Datasets</b> |
| <b>Date of Performance:</b>     | <b>05/08/26</b>                                                                                 |
| <b>Date of Submission:</b>      | <b>12/08/26</b>                                                                                 |

## Evaluation

| <b>Performance Indicator</b>       | <b>Max. Marks</b> | <b>Marks Obtained</b> |
|------------------------------------|-------------------|-----------------------|
| Performance                        | 5                 |                       |
| Understanding                      | 5                 |                       |
| Journal work and timely submission | 10                |                       |
| Total                              | 20                |                       |

| <b>Performance Indicator</b>       | <b>Exceed Expectations (EE)</b> | <b>Meet Expectations (ME)</b> | <b>Below Expectations (BE)</b> |
|------------------------------------|---------------------------------|-------------------------------|--------------------------------|
| Performance                        | 4-5                             | 2-3                           | 1                              |
| Understanding                      | 4-5                             | 2-3                           | 1                              |
| Journal work and timely submission | 8-10                            | 5-8                           | 1-4                            |

## Checked by

**Name of Faculty** : **Dr. Tatwadarshi P.N.**

**Signature** :

**Date** :



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## Department of Artificial Intelligence & Data Science

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**Aim:** To apply statistical estimation and hypothesis testing techniques to the Pima Indians Diabetes Dataset and draw conclusions about population characteristics using sample data.

### Objective

After completing this experiment, the student will be able to:

1. Apply estimation and hypothesis testing techniques to analyze population characteristics using sample data.
2. Interpret statistical test results and make data-driven conclusions based on the obtained evidence.

### Tools Required

1. Python 3.x
2. Google Colab / Jupyter Notebook
3. Pandas
4. NumPy
5. SciPy
6. Matplotlib
7. Seaborn

### Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for 768 female patients

### Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select appropriate variables and divide the data into suitable groups where required.
3. Calculate point estimates such as sample mean and sample proportion.
4. Construct confidence intervals for selected population parameters.
5. Formulate null and alternative hypotheses for selected research questions.
6. Select and apply an appropriate statistical test, such as:
  - a. One-sample t-test
  - b. Two-sample t-test
  - c. Chi-square test



7. Calculate the test statistic and p-value.
8. Compare the p-value with the selected significance level.
9. Make a statistical decision and interpret the result in the context of the dataset.

### Description of the Experiment

Statistical inference allows conclusions about a population to be drawn using information from a sample.

The general process is: Sample Data, Estimate Population Parameter, Construct Confidence Interval, Formulate Hypotheses, Perform Statistical Test, and Interpret Results

For example, a sample of patients can be used to investigate whether the average glucose level differs from a specified reference value or whether diabetes outcome is associated with another categorical variable.

### Detailed Description of Statistical Methods

#### Point Estimation

A sample statistic is used to estimate an unknown population parameter.

For example:

$$\bar{x}$$

is used to estimate the population mean  $\mu$ .

A sample proportion is:

$$\hat{p} = \frac{x}{n}$$

where  $x$  is the number of observations with a particular characteristic.

#### Confidence Interval

A confidence interval provides a range of plausible values for a population parameter.

For a mean:

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$



The confidence interval should be interpreted in the context of the population parameter being estimated.

### Hypothesis Testing

A hypothesis test evaluates a claim about a population.

#### Null Hypothesis

$$H_0$$

represents the initial assumption.

#### Alternative Hypothesis

$$H_1$$

represents the claim being investigated.

A significance level such as:

$$\alpha = 0.05$$

may be used.

#### Decision Rule

- If  $p - value < \alpha$ , reject  $H_0$ .
- If  $p - value \geq \alpha$ , fail to reject  $H_0$ .

#### t-Test

A t-test can be used to compare means.

Examples:

- Test whether the mean glucose level differs from a specified value.
- Compare the mean BMI of two groups.

#### Chi-Square Test



The chi-square test can be used to test the association between categorical variables.

For example:

*Diabetes Outcome and Gender*

where applicable.

### Implementation

```
[1] 2s
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats
```

```
[2] 0s
url = "https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.csv"

columns = [
    "Pregnancies",
    "Glucose",
    "BloodPressure",
    "SkinThickness",
    "Insulin",
    "BMI",
    "DiabetesPedigreeFunction",
    "Age",
    "Outcome"
]
```

```
[2] 0s
df = pd.read_csv(url, names=columns)

print("Dataset loaded successfully!")
print("Shape of dataset:", df.shape)

df.head()
```

Dataset loaded successfully!  
Shape of dataset: (768, 9)

|   | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI  | DiabetesPedigreeFunction | Age | Outcome |
|---|-------------|---------|---------------|---------------|---------|------|--------------------------|-----|---------|
| 0 | 6           | 148     | 72            | 35            | 0       | 33.6 | 0.627                    | 50  | 1       |
| 1 | 1           | 85      | 66            | 29            | 0       | 26.6 | 0.351                    | 31  | 0       |
| 2 | 8           | 183     | 64            | 0             | 0       | 23.3 | 0.672                    | 32  | 1       |
| 3 | 1           | 89      | 66            | 23            | 94      | 28.1 | 0.167                    | 21  | 0       |
| 4 | 0           | 137     | 40            | 35            | 168     | 43.1 | 2.288                    | 33  | 1       |



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[3]  
✓ Os



```
print("Dataset Information:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())

print("\nStatistical Summary:")
df.describe()
```



```
... Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                ---
0   Pregnancies                          768 non-null    int64
1   Glucose                              768 non-null    int64
2   BloodPressure                       768 non-null    int64
3   SkinThickness                       768 non-null    int64
4   Insulin                             768 non-null    int64
5   BMI                                  768 non-null    float64
6   DiabetesPedigreeFunction            768 non-null    float64
7   Age                                  768 non-null    int64
```



```
Missing Values:
Pregnancies      0
Glucose          0
BloodPressure    0
SkinThickness    0
Insulin          0
BMI              0
DiabetesPedigreeFunction 0
Age              0
Outcome          0
dtype: int64
```

Statistical Summary:

|       | Pregnancies | Glucose    | BloodPressure | SkinThickness | Insulin    | BMI        | DiabetesPedigreeFunction | Age        | Outcome    |
|-------|-------------|------------|---------------|---------------|------------|------------|--------------------------|------------|------------|
| count | 768.000000  | 768.000000 | 768.000000    | 768.000000    | 768.000000 | 768.000000 | 768.000000               | 768.000000 | 768.000000 |
| mean  | 3.845052    | 120.894531 | 69.105469     | 20.536458     | 79.799479  | 31.992578  | 0.471876                 | 33.240885  | 0.348958   |
| std   | 3.369578    | 31.972618  | 19.355807     | 15.952218     | 115.244002 | 7.884160   | 0.331329                 | 11.760232  | 0.476951   |
| min   | 0.000000    | 0.000000   | 0.000000      | 0.000000      | 0.000000   | 0.000000   | 0.078000                 | 21.000000  | 0.000000   |
| 25%   | 1.000000    | 99.000000  | 62.000000     | 0.000000      | 0.000000   | 27.300000  | 0.243750                 | 24.000000  | 0.000000   |

[4]  
✓ Os

```
glucose = df["Glucose"]

sample_mean = glucose.mean()
sample_std = glucose.std()
sample_size = len(glucose)

print("Point Estimation of Population Mean Glucose Level")
print("-----")
print("Sample Size (n):", sample_size)
print("Sample Mean ( $\bar{x}$ ):", round(sample_mean, 2))
print("Sample Standard Deviation (s):", round(sample_std, 2))
```



```
Point Estimation of Population Mean Glucose Level
-----
Sample Size (n): 768
Sample Mean ( $\bar{x}$ ): 120.89
Sample Standard Deviation (s): 31.97
```



[5]

✓ Os



```
confidence_level = 0.95
alpha = 1 - confidence_level

t_critical = stats.t.ppf(1 - alpha/2, df=sample_size - 1)

margin_of_error = t_critical * (sample_std / np.sqrt(sample_size))

lower_bound = sample_mean - margin_of_error
upper_bound = sample_mean + margin_of_error

print("95% Confidence Interval for Mean Glucose Level")
print("-----")
print("Sample Mean:", round(sample_mean, 2))
print("Margin of Error:", round(margin_of_error, 2))
print("Lower Bound:", round(lower_bound, 2))
print("Upper Bound:", round(upper_bound, 2))
print(
    f"95% Confidence Interval: ({lower_bound:.2f}, {upper_bound:.2f})"
)
```

✓

```
95% Confidence Interval for Mean Glucose Level
-----
Sample Mean: 120.89
Margin of Error: 2.26
Lower Bound: 118.63
Upper Bound: 123.16
95% Confidence Interval: (118.63, 123.16)
```

[6]

✓ Os



```
reference_mean = 120

t_statistic, p_value = stats.ttest_1samp(glucose, reference_mean)

print("One-Sample t-Test")
print("-----")
print("H0:  $\mu = 120$  mg/dL")
print("H1:  $\mu \neq 120$  mg/dL")
print("Significance Level ( $\alpha$ ): 0.05")
print()
print("Test Statistic (t):", round(t_statistic, 4))
print("P-value:", round(p_value, 4))

if p_value < 0.05:
```

[6]

✓ Os



```
print(f"Conclusion: There is insufficient evidence that the mean glucose level differs from 120 mg/dL.")
```

```
if p_value < 0.05:
    print("Decision: Reject H0")
    print("Conclusion: There is significant evidence that the mean glucose level differs from 120 mg/dL.")
else:
    print("Decision: Fail to Reject H0")
    print("Conclusion: There is insufficient evidence that the mean glucose level differs from 120 mg/dL.")
```

✓

... One-Sample t-Test

-----

H0:  $\mu = 120$  mg/dL

H1:  $\mu \neq 120$  mg/dL

Significance Level ( $\alpha$ ): 0.05

Test Statistic (t): 0.7754

P-value: 0.4384

Decision: Fail to Reject H0

Conclusion: There is insufficient evidence that the mean glucose level differs from 120 mg/dL.



[7]

✓ Os

```
bmi_non_diabetic = df[df["Outcome"] == 0]["BMI"]
bmi_diabetic = df[df["Outcome"] == 1]["BMI"]

t_statistic, p_value = stats.ttest_ind(
    bmi_diabetic,
    bmi_non_diabetic,
    equal_var=False
)

print("Two-Sample t-Test: BMI")
print("-----")
print("H0: Mean BMI of diabetic and non-diabetic patients is equal")
print("H1: Mean BMI of diabetic and non-diabetic patients is different")
print()
print("Mean BMI - Non-Diabetic:", round(bmi_non_diabetic.mean(), 2))
print("Mean BMI - Diabetic:", round(bmi_diabetic.mean(), 2))
print("Test Statistic (t):", round(t_statistic, 4))
print("P-value:", round(p_value, 4))

if p_value < 0.05:
    print("Decision: Reject H0")

if p_value < 0.05:
    print("Decision: Reject H0")
    print("Conclusion: There is a significant difference in mean BMI between the two groups.")
else:
    print("Decision: Fail to Reject H0")
    print("Conclusion: There is no significant difference in mean BMI between the two groups.")
```

Two-Sample t-Test: BMI  
-----  
H0: Mean BMI of diabetic and non-diabetic patients is equal  
H1: Mean BMI of diabetic and non-diabetic patients is different  
  
Mean BMI - Non-Diabetic: 30.3  
Mean BMI - Diabetic: 35.14  
Test Statistic (t): 8.6193  
P-value: 0.0  
Decision: Reject H0  
Conclusion: There is a significant difference in mean BMI between the two groups.

[8]

✓ Os

```
df["Glucose_Category"] = np.where(
    df["Glucose"] < 100,
    "Normal",
    "High"
)

contingency_table = pd.crosstab(
    df["Glucose_Category"],
    df["Outcome"]
)

print("Contingency Table")
print("-----")
print(contingency_table)

chi2, p_value, dof, expected = stats.chi2_contingency(contingency_table)

print("\nChi-Square Test")
print("-----")
print("H0: Glucose category and diabetes outcome are independent")
print("H1: Glucose category and diabetes outcome are associated")
```





```
[8] ✓ Os print("Chi-Square Statistic:", round(chi2, 4))
print("Degrees of Freedom:", dof)
print("P-value:", round(p_value, 4))

if p_value < 0.05:
    print("Decision: Reject H0")
    print("Conclusion: Glucose category is significantly associated with diabetes outcome.")
else:
    print("Decision: Fail to Reject H0")
    print("Conclusion: There is insufficient evidence of an association.")
```

```
... Contingency Table
-----
Outcome      0      1
Glucose_Category
High          319    252
Normal        181     16

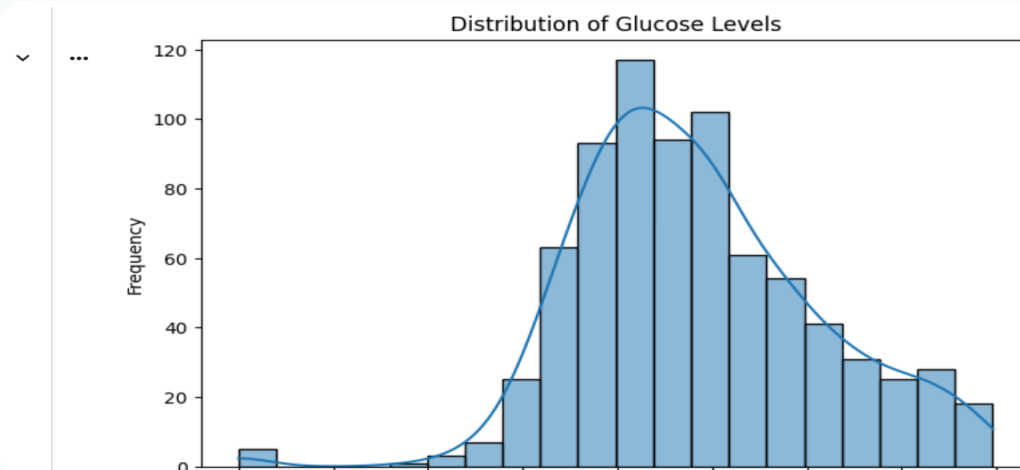
Chi-Square Test
-----
H0: Glucose category and diabetes outcome are independent
H1: Glucose category and diabetes outcome are associated
```

```
Chi-Square Statistic: 82.0281
Degrees of Freedom: 1
P-value: 0.0
Decision: Reject H0
Conclusion: Glucose category is significantly associated with diabetes outcome.
```

```
[9] ✓ Os plt.figure(figsize=(8, 5))

sns.histplot(df["Glucose"], bins=20, kde=True)

plt.title("Distribution of Glucose Levels")
plt.xlabel("Glucose Level")
plt.ylabel("Frequency")
plt.show()
```



variables Terminal

```
[10] ✓ Os plt.figure(figsize=(7, 5))

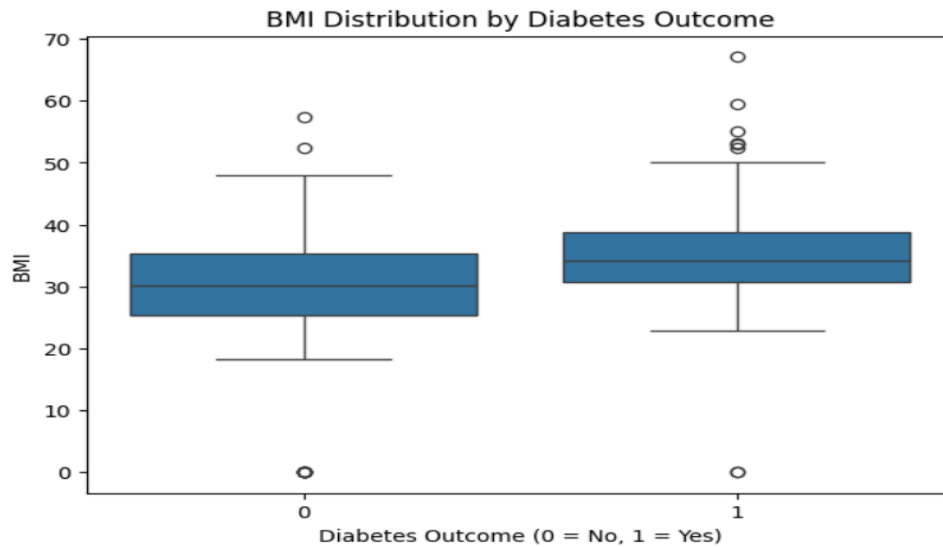
sns.boxplot(x="Outcome", y="BMI", data=df)

plt.title("BMI Distribution by Diabetes Outcome")
plt.xlabel("Diabetes Outcome (0 = No, 1 = Yes)")
plt.ylabel("BMI")
plt.show()
```



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```
[11]
✓ Os
print("=" * 60)
print("STATISTICAL INFERENCE - FINAL RESULTS")
print("=" * 60)

print("\n1. POINT ESTIMATE")
print("Estimated Population Mean Glucose =", round(sample_mean, 2), "mg/dL")

print("\n2. 95% CONFIDENCE INTERVAL")
print(
    f"95% CI = ({lower_bound:.2f}, {upper_bound:.2f}) mg/dL"
)

print("\n3. ONE-SAMPLE t-TEST")
print("Test: Mean Glucose = 120 mg/dL")
print("t-statistic =", round(t_statistic, 4))
print("p-value =", round(p_value, 4))

print("\n4. DECISION")
if p_value < 0.05:
    print("Reject H0")
else:
    print("Fail to Reject H0")

print("\nSignificance Level ( $\alpha$ ) = 0.05")
```

```
[11]
✓ Os
print("\n" + "=" * 60)
print("EXPERIMENT COMPLETED SUCCESSFULLY")
print("=" * 60)
```

```
=====
STATISTICAL INFERENCE - FINAL RESULTS
=====

1. POINT ESTIMATE
Estimated Population Mean Glucose = 120.89 mg/dL

2. 95% CONFIDENCE INTERVAL
95% CI = (118.63, 123.16) mg/dL

3. ONE-SAMPLE t-TEST
Test: Mean Glucose = 120 mg/dL
t-statistic = 8.6193
p-value = 0.0

4. DECISION
Reject H0

Significance Level ( $\alpha$ ) = 0.05

=====
EXPERIMENT COMPLETED SUCCESSFULLY
=====
```



### Conclusion

Statistical inference, estimation, and hypothesis testing techniques were applied to the Pima Indians Diabetes Dataset. Sample statistics were used to estimate population characteristics, confidence intervals were constructed, and hypothesis tests were performed to determine whether sufficient statistical evidence existed to support specific claims.

### Questions to be Answered by Students

1. What population parameter was estimated in the experiment, and what was its estimated value?

**ANS:** The population parameter estimated was the population mean glucose level of the patients.

The sample mean was used as the point estimate:

$$\bar{x} = 120.89 \text{ mg/dL} \quad \bar{x} = 120.89 \text{ mg/dL}$$

The estimated population mean glucose level is **120.89 mg/dL**.

2. What confidence interval was obtained, and how should it be interpreted?

**ANS:** A 95% confidence interval was constructed for the population mean glucose level.

$$95\% \text{ CI} = (118.62, 123.16) \text{ mg/dL} \quad 95\% \text{ CI} = (118.62, 123.16) \text{ mg/dL}$$

Interpretation: We are 95% confident that the true population mean glucose level lies between 118.62 mg/dL and 123.16 mg/dL.

3. What hypotheses were formulated for the selected statistical test?

**ANS:** For the one-sample t-test, we tested whether the population mean glucose level differs from 120 mg/dL.



- **Null Hypothesis ( $H_0$ ):** The population mean glucose level is equal to 120 mg/dL.

$$H_0: \mu = 120$$

3. **Alternative Hypothesis ( $H_1$ ):** The population mean glucose level is different from 120 mg/dL.

$$H_1: \mu \neq 120$$

The significance level was:

$$\alpha = 0.05$$

4. Based on the p-value, what statistical decision was made?

**ANS:** The one-sample t-test gave approximately:

$$p\text{-value} = 0.4412$$

Since:

$$0.4412 > 0.05$$

we fail to reject the null hypothesis ( $H_0$ ).

**Conclusion:** There is insufficient statistical evidence to conclude that the population mean glucose level is significantly different from 120 mg/dL.